Beyond Benchmark Accuracy: Auditing LLM Reliability under Urdu Linguistic Variation

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Abstract

Large Language Models (LLMs) perform well on many Natural Language Processing (NLP) tasks, but standard benchmarks only partly reflect their reliability in low-resource languages like Urdu. Urdu appears in diverse written forms, including standard Urdu script, Roman Urdu, Urdu–English code-mixing, and noisy or informal user text, which can trigger inconsistent model behavior without changing the underlying meaning. This framework evaluates modern LLMs under such variation in task performance, semantic correctness, robustness, and behavioral consistency. Additionally, it incorporates an LLM-as-a-Judge evaluation to assess semantic correctness in Urdu question answering (QA). Answer consistency is measured with the Semantic Consistency Score (SCS), and explanation consistency with the Reasoning Consistency Score (RCS). Six general-purpose LLMs (three commercial and three open-weight) are tested under multiple Urdu conditions. The results show that standard task metrics can miss important robustness and consistency gaps, underscoring the need to evaluate LLMs under realistic linguistic variation before a reliability-critical deployment.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; **Natural language processing**.

Keywords

Large Language Models, Urdu NLP, LLM Evaluation, Linguistic Variation, Behavioral Reliability, Semantic Consistency, Culturally Grounded Reasoning Evaluation, Cross-cultural Evaluation

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CIKM, Rome, Italy
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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/XXXXXXX.XXXXXXX

ACM Reference Format:
Muhammad Rehman Zafar, Hina Tariq, Mahwish Shahid, and Fareeha Anwar. 2026. Beyond Benchmark Accuracy: Auditing LLM Reliability under Urdu Linguistic Variation. In *REACH-AI Workshop at CIKM 2026*. ACM, New York, NY, USA, 7 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Large Language Models (LLMs) demonstrate strong performance in a wide range of Natural Language Processing (NLP) tasks, including answering questions, summarizing, machine translation, and information extraction[1–4]. As these models are increasingly applied across diverse linguistic settings, evaluating their reliability under realistic input conditions has become as important as measuring their accuracy. Conventional benchmarks typically assess downstream tasks using metrics such as accuracy, F1-score, BLEU, ROUGE, and Exact Match (EM) [5–7]. Although useful for standardized comparison, these metrics capture only part of LLMs behavior and may overlook semantic correctness, robustness, and consistency.

Behavioral and holistic assessments indicate that strong benchmark performance does not necessarily ensure reliable behavior under realistic input variations: minor, meaning-preserving edits can alter model outputs [8, 9]. However, most robustness work targets English and other high-resource languages, leaving low-resource settings understudied. This gap matters for Urdu, which is frequently written in standard Urdu script, Roman Urdu, mixed Urdu-English script, and noisy or informal text. Figure 1 illustrates this variation using the same underlying question expressed in standard Urdu, Roman Urdu, and mixed Urdu-English script. Although these forms convey the same meaning, they may elicit inconsistent model responses. Meaning-equivalent forms can trigger different

Linguistic Characteristics	Examples for How many provinces does Pakistan have?
Standard Urdu	پاکستان کے کتنے صوبے ہیں؟
Roman Urdu	Pakistan ke kitne soobay hain?
Mixed Script	Pakistan ke kitne provinces hain?

Figure 1: Example of meaning-preserving linguistic variation across standard Urdu, Roman Urdu, and mixed script.

responses despite preserving the intended meaning, so standardized Urdu benchmarks may miss real-world reliability issues. In addition to that, generative tasks, such as answering questions, add further difficulty. EM, BLEU, and ROUGE depend on lexical overlap and can undervalue semantically correct expressions, especially in open-ended QA. LLM-as-a-Judge methods can evaluate beyond lexical metrics [10], but their use for Urdu QA and how they relate to robustness and consistency under linguistic variation remain unclear.

This study empirically evaluates contemporary general-purpose LLMs under Urdu linguistic variation. The framework assesses Urdu LLM reliability across task performance, semantic correctness, robustness to variation, answer-level semantic consistency, and explanation-level consistency. Six LLMs (three commercial, three open-weight) are tested on five Urdu NLP tasks: text classification, named entity recognition (NER), question answering (QA), summarization, and machine translation. For Urdu QA, an LLM-as-a-Judge is used to evaluate semantic correctness beyond exact lexical matching.

Furthermore, answer-level consistency is measured by the Semantic Consistency Score (SCS) and explanation-level consistency by the Reasoning Consistency Score (RCS), used as evaluation metrics rather than new similarity algorithms. SCS checks whether answers preserve meaning across meaning-equivalent variants, while RCS checks whether explanations remain semantically consistent across the same variants. The RCS quantifies the semantic consistency of articulated explanations across linguistic variants rather than the substantive validity of the underlying reasoning; consequently, a high RCS should not be interpreted as evidence of correct reasoning or internal reasoning consistency.

To guide the empirical investigation, the study addresses the following three research questions: i) Do conventional lexical evaluation metrics accurately reflect the semantic correctness of LLM-generated answers for Urdu question answering? ii) How does linguistic variation affect LLM task performance, answer-level semantic consistency, and explanation-level consistency? and iii) Do multidimensional reliability measures reveal model differences that conventional task metrics fail to capture?

The main contributions of this paper can be summarized as follows:

- A multidimensional empirical evaluation framework is presented to characterize the reliability of contemporary general-purpose LLMs in task performance, semantic correctness, robustness, and consistency under Urdu linguistic variation.
- A controlled evaluation protocol is established for comparing six general-purpose LLMs, comprising three commercial models and three open-weight models, across five Urdu NLP tasks and controlled linguistic variations.
- An LLM-as-a-Judge evaluation protocol is incorporated to assess the semantic correctness of Urdu question-answering responses, complementing conventional lexical metrics and enabling evaluation beyond surface-level word overlap.
- An empirical analysis demonstrates that conventional benchmark metrics alone may not fully capture important differences in the robustness and consistency of contemporary LLMs under Urdu linguistic variation.

2 Related Work

Recent advances in LLMs have accelerated the development of multilingual and language-specific evaluation benchmarks, including for low-resource languages such as Urdu. Early large-scale evaluations of pretrained LLMs for Urdu NLP identified performance gaps between multilingual LLMs and task-specific systems [15]. Subsequent work introduced benchmarks that target cultural and domain-specific capabilities, including native queries and cultural knowledge [16], historical NER [17], reasoning evaluation [18], educational knowledge assessment [19], and mental health classification [20]. While these efforts have substantially expanded Urdu NLP resources, they primarily focus on benchmark construction and downstream task performance rather than the reliability of LLM behavior under natural linguistic variation.

The general LLM evaluation has evolved from task-specific accuracy toward more comprehensive assessment. Existing frameworks evaluate dimensions such as accuracy, robustness, calibration, fairness, and efficiency [11], while standardized benchmarks assess knowledge and reasoning across multiple tasks [12, 13]. The LLM-as-a-Judge approaches have further enabled the evaluation of open responses [10, 14], with automatic methods designed to improve alignment with human judgments [21, 22]. Semantic evaluation metrics have also been proposed to measure similarity between generated and reference texts [26]. Research on self-consistency, hallucination, and reliable LLMs further highlights the importance of robustness and reliability of reasoning beyond benchmark accuracy [23–25].

Despite progress, previous work does not systematically assess whether LLM behavior remains consistent when the same underlying meaning is expressed in standard Urdu, Roman Urdu, Urdu-English code-mixed, and noisy informal forms; Table 1 summarizes related work. To address this gap, this study introduces complementary evaluation pipelines for task performance, robustness, and cross-variant behavioral consistency. SCS measures the semantic invariance at the answer-level, while RCS measures the semantic consistency of the model-generated explanations in equivalent Urdu inputs. These are task-specific consistency measures instantiated using multilingual sentence embeddings rather than new similarity algorithms.

3 Methodology

This section presents a unified evaluation framework, shown in Figure 2, to systematically evaluate LLMs for Urdu language in terms of task performance, quality of answers, robustness, and behavioral consistency. The framework consists of three complementary components. First, a multi-task evaluation pipeline benchmarks LLMs on core Urdu NLP tasks, including text classification, NER, QA, summarization, and machine translation. Robustness is evaluated separately using controlled word substitutions in sentiment items, while cross-variant consistency is evaluated over standard Urdu, Roman Urdu, Urdu–English code-mixed, and noisy Urdu forms. Second, an LLM-as-a-Judge (gpt-4o-mini) framework is used to evaluate generative QA, allowing the semantic assessment of the correctness of the answer beyond traditional lexical overlap metrics. Third, a consistency evaluation pipeline is introduced that measures model behavior across semantically equivalent Urdu variants using

Table 1: Comparison of existing LLM evaluation studies and their limitations with respect to Urdu reliability assessment.

Research Theme	Representative Studies	Limitations
General LLM Evaluation	HELM [11], MMLU [12], MMLU-Pro [13], MT-Bench [10], Survey of LLM Evaluation [14]	Primarily evaluates the task's overall performance, the logic applied, and the benchmark's accuracy. It does not assess semantic consistency and/or the stability of reasoning under linguistic variations.
Urdu LLM Benchmarks	Benchmarking Urdu LLMs [15], NativQA [16], QaLLM [17], UrduBench [18], UrduMMLU [19], Urdu Mental Health Corpus [20]	Focuses on developing benchmarks and evaluating downstream tasks. Gives only limited attention to the robustness of spelling variations, dialects, script conventions, Roman Urdu, or the consistency of generated outputs.
Automatic Evaluation	G-Eval [21], JudgeLM [22]	Provides consistent automated quality evaluation, but focuses on scoring individual responses rather than checking consistency across semantically equivalent inputs. LLM-based judges can also introduce language bias and evaluation bias.
Reliability and Trustworthiness	Self-Consistency [23], Hallucination Survey [24], Trustworthy LLMs [25]	The work covers concepts such as the reliability of reasoning, hallucinations, and trustworthy AI, but it does not address evaluation metrics for semantic invariance or reasoning consistency in low-resource languages such as Urdu.
Semantic Evaluation	BLEURT [26]	Calculates the semantic similarity between the generated and reference text, but does not assess semantic consistency across different responses produced from similar linguistic forms.

two metrics: the SCS, which quantifies semantic preservation across variants, and the RCS, which evaluates the semantic consistency of model-generated explanations across variants. Together, these components provide a comprehensive, reproducible, and model-agnostic evaluation methodology that supports the development and comparison of reliable Urdu LLMs.

4 Experimental Design and Results

This section begins by outlining how the models were assessed, covering the datasets, experimental design, evaluation metrics, and reliability checks. It then reports the findings, comparing performance across tasks, QA scoring approaches, and consistency among different Urdu writing styles and task types.

4.1 Evaluation Setup

Six LLMs are evaluated for sentiment classification (60 items), NER (40), QA (40 unique prompts, each processed in two repeated runs), summarization (20) and bidirectional translation (40). Consistency is evaluated using 70 items written in standard Urdu, Roman Urdu, code mixed Urdu, and noisy Urdu. This produces 280 inputs and 210 comparisons per model. The models are accessed through OpenRouter with temperature 0. Answers and explanations are generated separately. Semantic similarity is measured using multilingual MiniLM L12 v2.

4.2 Evaluation Metrics

Evaluation protocols were selected to align with the structural properties of each task's outputs. Sentiment classification performance was quantified using accuracy and macro-averaged F1, while NER was evaluated via entity-type micro-averaged F1. For open-ended text generation, summarization and translation, performance was evaluated via cosine similarity between the generated and reference representations, so that semantically equivalent responses received credit even when they differed lexically. QA was evaluated using three complementary metrics to characterize the limitations of purely lexical matching; EM captures strict string-level equivalence, whitespace-set F1 (WS-F1) measures token-overlap at the set level, and QA-J employs GPT-4o-mini as a large language model (LLM) judge to estimate semantic correctness. The responses assigned a QA-J rating of 4 or 5 on a five-point scale were considered correct.

4.3 Consistency Evaluation

Consistency measures whether a model preserves the meaning of its output when the same input is expressed in different forms of Urdu. Each of the 70 standard Urdu items is paired with three variants: Roman Urdu, code mixed Urdu, and noisy Urdu. In this evaluation, two complementary scores are used. The SCS measures the similarity between the generated answers, whereas the RCS measures the semantic similarity between the generated explanations across the same variants.

Let $\mathcal{I}$ denote the set of evaluation items and $\mathcal{V}$ the set of linguistic variants. For model m and item i, $a_{m,i}^{(0)}$, and $r_{m,i}^{(0)}$ represent the answer and explanation generated from standard Urdu. The corresponding outputs for variant v are denoted by $a_{m,i}^{(v)}$ and $r_{m,i}^{(v)}$. The encoder ϕ maps each output to a multilingual sentence embedding.

The item-level scores are calculated as

$$\begin{aligned} \mathrm{SCS}_{m,i}^{(v)} &= \cos\Big(\phi(a_{m,i}^{(0)}), \phi(a_{m,i}^{(v)})\Big), \\ \mathrm{RCS}_{m,i}^{(v)} &= \cos\Big(\phi(r_{m,i}^{(0)}), \phi(r_{m,i}^{(v)})\Big). \end{aligned} \tag{1}$$

The overall scores for each model are obtained by averaging across all items and variants:

$$\begin{aligned} \mathrm{SCS}_m &= \frac{1}{|\mathcal{I}||\mathcal{V}|} \sum_{i\in\mathcal{I}} \sum_{v\in\mathcal{V}} \mathrm{SCS}_{m,i}^{(v)}, \\ \mathrm{RCS}_m &= \frac{1}{|\mathcal{I}||\mathcal{V}|} \sum_{i\in\mathcal{I}} \sum_{v\in\mathcal{V}} \mathrm{RCS}_{m,i}^{(v)}. \end{aligned} \tag{2}$$

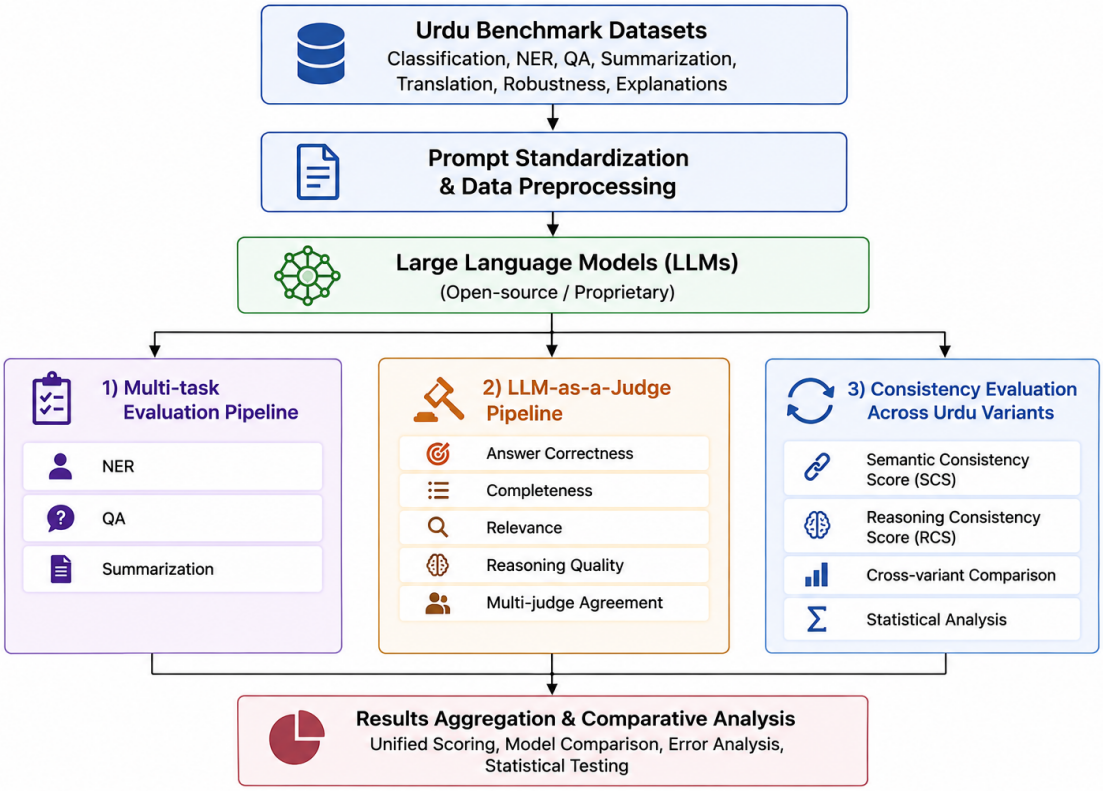


Figure 2: Overview of the proposed Urdu LLM evaluation framework.

Table 2: Performance on standard Urdu tasks. QA-J is the proportion of responses judged correct. Summarization and translation use cosine similarity. QA scores combine two repeated runs of 40 unique prompts.

Model	Sent. Acc.	Sent. F1	NER F1	QA WS-F1	QA-J	Sum.	Trans.
GPT-4o	0.800	0.791	**0.692**	0.326	1.000	**0.854**	**0.918**
Claude	**0.900**	**0.897**	0.658	0.309	1.000	0.836	0.905
Gemini	0.867	0.863	0.689	**0.723**	1.000	0.801	0.772
Ministral	0.733	0.731	0.571	0.085	0.938	0.716	0.583
Qwen	0.683	0.683	0.510	0.349	1.000	0.818	0.673
Llama-3.2-3B	0.467	0.356	0.395	0.403	0.775	0.795	0.712

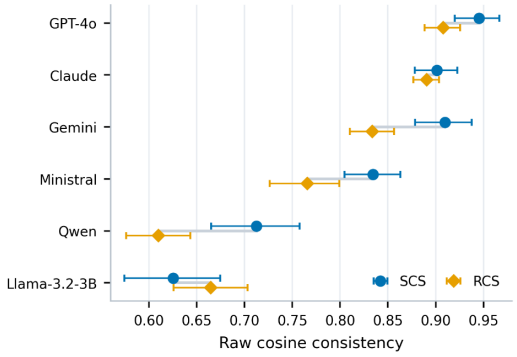
Since $|\mathcal{I}| = 70$ and $|\mathcal{V}| = 3$, each overall score summarizes 210 comparisons. Values closer to 1 indicate that the model produces more similar output in Urdu forms.

SCS and RCS measure consistency rather than correctness. A high SCS may reflect answers that are consistently similar but incorrect. Similarly, a high RCS may reflect explanations that are consistently repeated but incorrect or unsupported. RCS measures semantic similarity in the expressed explanations and does not evaluate the validity of the reasoning or reveal the model's internal reasoning process. Therefore, both scores should be interpreted alongside task performance and human evaluation.

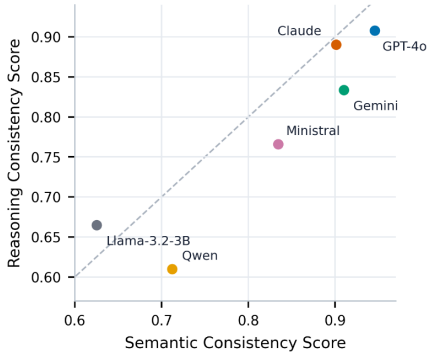
Robustness is evaluated separately using controlled word substitutions in 60 sentiment items. Explanation stability is calculated as the mean pairwise cosine similarity among three repeated explanations for each QA prompt. The 95% confidence intervals are estimated using 10,000 item-level bootstrap resamples, with all variants of the same item resampled together.

4.4 Task Performance

No single model performed best on all tasks. Claude achieved the highest sentiment accuracy (0.900), while GPT-4o performed best in NER (0.692), summarization (0.854) and translation (0.918). The complete results are reported in Table 2 and visualized in Figure 4.



(a) Overall SCS and RCS with 95% bootstrap confidence intervals.



(b) Relationship between SCS and RCS.

Figure 3: Answer- and explanation-level consistency across the evaluated models.

4.5 QA Metric Comparison

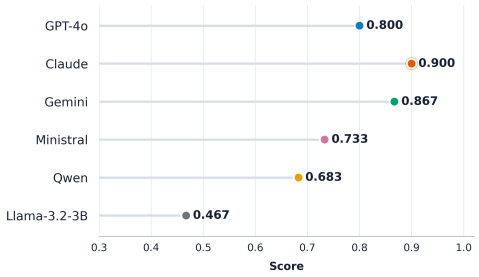
Figure 5 compares the three QA metrics. GPT-4o did not receive exact matches and a WS-F1 score of 0.326, but the LLM judge accepted all its answers as correct. Claude and Qwen showed a similar pattern. Gemini achieved the strongest lexical scores, with EM of 0.563 and WS-F1 of 0.723.

These results show that exact-match measures, such as EM, can undervalue performance by penalizing answers that are semantically correct but phrased differently, underscoring the shortcomings of lexical metrics for assessing generative Urdu QA. However, human validation is necessary to confirm the trustworthiness of LLM-based judgments, highlighting a human-in-the-loop approach to ensure that these evaluations are reliable.

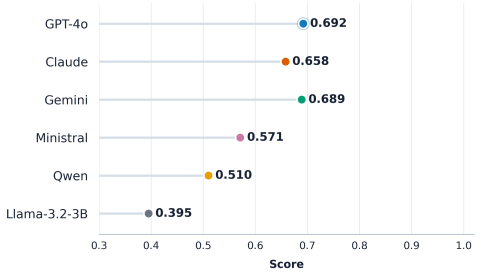
4.6 Consistency and Robustness

Figure 3 compares the consistency of the answer and the explanation. GPT-4o achieved the highest SCS (0.945) and RCS (0.908). Qwen and Gemini showed the largest differences between the two scores, indicating that stable answers do not always produce stable explanations.

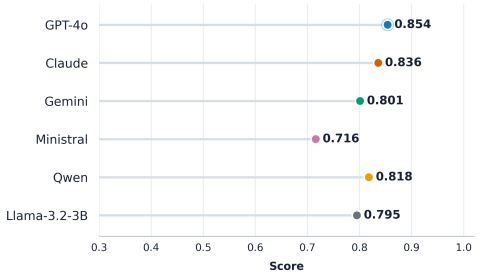
Figure 6 shows the consistency between Urdu forms and tasks. Noisy Urdu produced the lowest SCS for four of the six models. The differences in tasks were also visible. For example, Qwen's



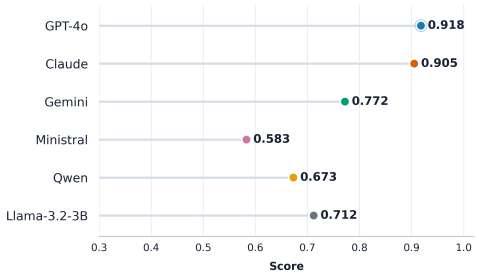
(a) Sentiment accuracy.



(b) NER entity-type micro-F1.



(c) Summarization cosine similarity.



(d) Translation cosine similarity.

Figure 4: Performance across four standard Urdu tasks. Higher values indicate better performance.

SCS decreased from 0.826 for sentiment to 0.583 for summarization, while QA produced the highest RCS for every model.

In the sentiment robustness test, Claude achieved the highest accuracy (0.900), while Llama-3.2-3B achieved the lowest (0.479). The explanation stability ranged from 0.887 for Ministral to 0.968 for

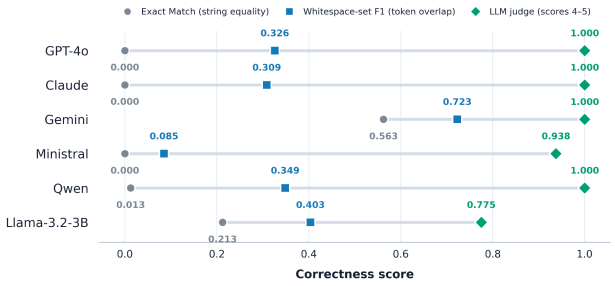


Figure 5: Comparison of exact matching, word overlap, and LLM judged correctness for Urdu QA.

Gemini. These findings show that consistency should be considered together with correctness because a model can repeat the same incorrect explanation.

5 Limitations

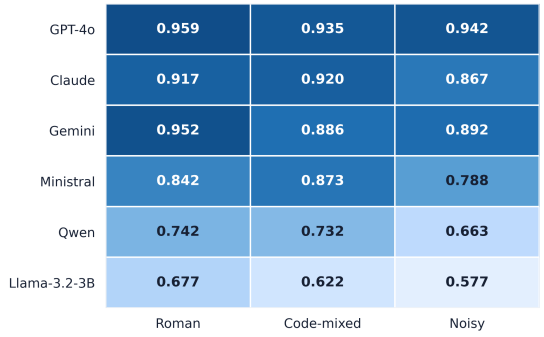
This study examines only a particular subset of Urdu tasks, linguistic varieties, and LLMs; therefore, the findings may have limited external validity and may not extend to every application area or to Urdu dialects that were not directly analyzed. Since SCS and RCS capture similarity rather than actual task performance, they should be considered alongside task-level accuracy metrics. Future research will broaden the supporting datasets and add more comprehensive human evaluation procedures.

6 Conclusion and Future Work

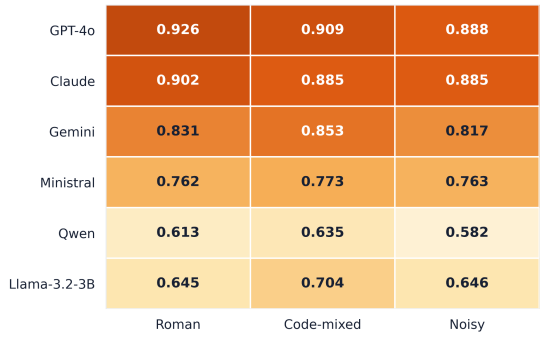
The results show that no single metric can fully capture Urdu LLM performance. Claude achieved the highest sentiment accuracy at 90.0%, while GPT-4o achieved the highest scores for NER (F1 = 0.692), summarization (0.854) and translation (0.918). For QA, GPT-4o received an Exact Match score of 0% and a WS-F1 score of 32.6%, although the LLM judge rated all its answers as semantically correct. This result shows that lexical metrics may underestimate the correct answers expressed using different words.

The consistency analysis provided further information about the reliability of the model. GPT-4o achieved the highest SCS (0.945) and RCS (0.908). Noisy Urdu reduced the consistency of the responses for four of the six models, while smaller models were generally more sensitive to linguistic variation. These findings show that consistent answers do not always produce consistent explanations.

Future work will extend the benchmark to explicitly include the dialectal and regional variation of Urdu, additional domains, cultural tasks, and language models. It will also include a broader human evaluation and compare different LLM judges and embedding methods.



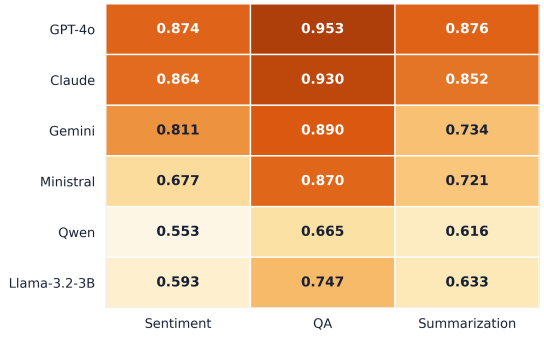
(a) SCS by linguistic variant.



(b) RCS by linguistic variant.



(c) SCS by task.



(d) RCS by task.

Figure 6: Answer- and explanation-level consistency across Urdu linguistic variants and tasks. Darker cells indicate greater consistency.

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